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ZUCTI

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Lab 1 — AI-Driven Detection

Choosing a SOC classifier and finding attacks without labels on CSE-CIC-IDS2018

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Repository	github.com/MZU-GH/SEC-635-Lab-1

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Executive summary

Two SOC problems on CSE-CIC-IDS2018 [1], on Wednesday brute force (FTP, SSH) and Thursday infiltration: which classifier to deploy (Task 1), and whether attacks can be found without labels (Task 2). All splits are time-ordered on a repaired clock.

- **Task 1 answer: Random Forest (RF) for attack families with labelled history.** This model raises **49** false alerts per day (95% CI 18–107) on a brute-force campaign it has seen, and that is acceptable because it costs about 0.8 analyst-hours per day at 1 min per alert (10% of one 8-hour shift) for 99.99% recall. On Thursday infiltration the same model raises 131,693 false alerts per day, which is not acceptable (Section 5).
- **ML does not catch an unseen attack type; a rule does.** On Wed 70% the SSH attack was absent from training and RF recall was 0.0002, while the port 21/22 rule catches all of it (recall 1.000). The rule was written knowing the ports, so this favours it.
- **Clustering separates brute force without labels.** DBSCAN reaches F1 0.9994 on Wednesday for 700 analyst labels. On infiltration all three detectors miss 70% of the attack flows and a one-line zero-payload rule matches their F1 (0.387 against 0.388). The only finds no rule reproduces are the GMM-only (85) and DBSCAN-only (97) long, near-silent connections.
- **Caveats.** The 49/day rests on 6 false positives, and Wed 80% shares one SSH campaign between training and test (an optimistic bound). A random split inflates Thursday RF F1 by only 0.102; the unrepaired clock distorts far more.
- **The ceiling.** 40% of sampled infiltration flows are feature-identical to some benign flow, so flow-level detection of infiltration is capped.

1 Methodology

1.1 Data and cleaning

Both days were cleaned in the same order: repeated header lines, rows dated 1970 and rows holding ±Infinity were removed, then constant columns were dropped (Table 1). NaNs occurred only in rows that also held Infinity (2,277 and 1,834 rows). Duplicate flows were counted and kept: identical flows are legitimate repeated traffic. *Timestamp* and *Dst Port* were dropped as features (time reveals the attack window; ports 21 and 22 almost define the Wednesday labels); *Protocol* was kept. The label is binary. Thursday has no 1970 rows.

Table 1 — Data cleaning and class balance per day. Rows removed add up exactly to rows in file minus rows kept.

Step	Wednesday 14-02-2018	Thursday 01-03-2018
Rows in file	1,048,575	331,125
Removed: repeated header lines / rows dated 1970 / rows with ±Infinity	0 / 5 / 3,824	25 / 0 / 2,919
Rows kept	1,044,746	328,181
Constant columns dropped / features used	10 / 67	8 / 69
Duplicate rows (counted, kept)	373,613	40,154
Attack flows (share of rows kept)	193,354 FTP + 187,589 SSH (36.5%)	92,403 Infiltration (28.2%)

1.2 Clock repair and the honest split

The raw hour histogram holds traffic in hours 01–05 and 08–12 and none in hours 06–07, which fits a 12-hour clock written without AM/PM (the afternoon is stored as 01–05). SSH brute force appears at raw 02:01–03:32 and becomes 14:01–15:32 after adding 12 hours. The repair moved 377,414 Wednesday and 163,781 Thursday flows and leaves one daytime window per day (9.0 h and 9.5 h). The evidence is data-internal only: the UNB schedule page could not be fetched, so the published schedule is not verified, and file order is not chronological (289,263 backward steps on Wednesday).

The first 70% of flows in time train and the last 30% test (Figure 1, Table 2). Wed 70% tests SSH, never seen in training, so it measures unseen-attack behaviour, not leakage. We added Wed 80% after seeing that result, to test a known attack arriving later; its training and test windows share one SSH campaign, so it is an optimistic bound. Thu 70% cuts the second infiltration wave. Table 4 shows why the split matters.

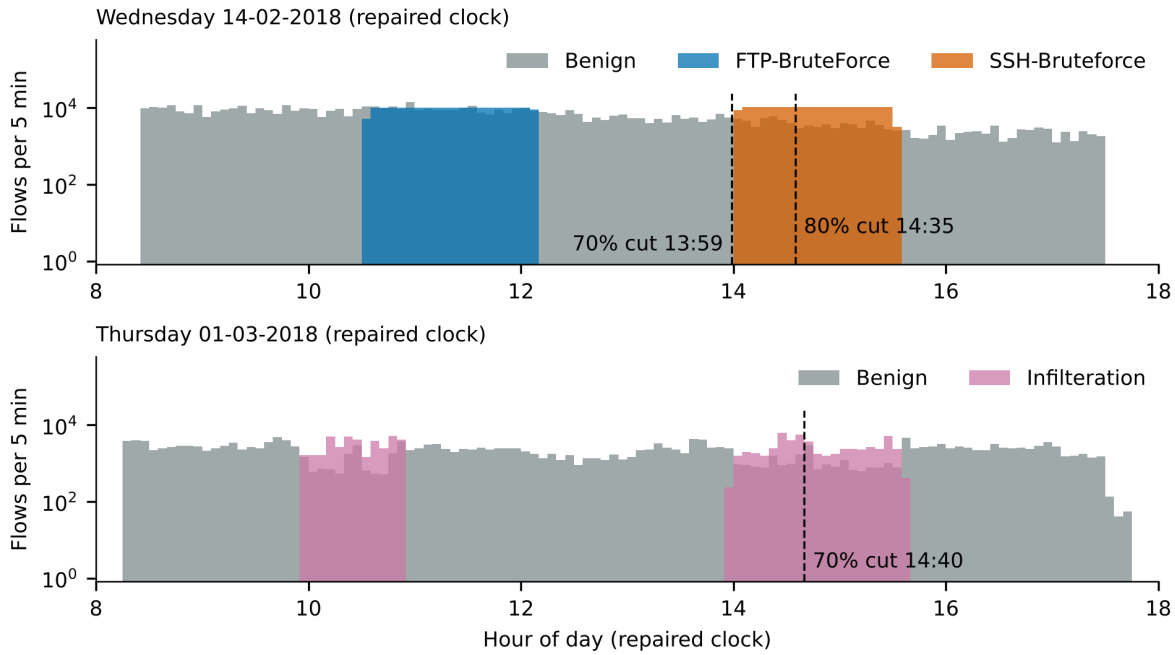


Figure 1 — The repaired clock gives each attack a compact window, and the cuts fall inside them. Flows per 5 min by label (log scale) after the clock fix, with the 70% cut (Wed 70%, 13:59), the 80% cut (Wed 80%, 14:35) and the 70% cut (Thu 70%, 14:40). Top: Wednesday, where FTP trains and SSH tests. Bottom: Thursday, two infiltration waves.

Table 2 — Time-ordered splits. Validation window = last 20% of the training rows, used for tuning only.

Split	Cut	Training attacks	Test attacks	Test window (h)	Validation-window attacks
Wed 70%	13:59	193,354 FTP	187,589 SSH (unseen)	3.5	10,683 FTP
Wed 80%	14:35	193,354 FTP + 69,794 SSH	117,795 SSH (seen)	2.9	69,794 SSH (not in fit part)
Thu 70%	14:40	63,345 Infilteration	29,058 Infilteration	3.1	24,808 Infilteration

1.3 Task 1 configuration and evaluation

Models and preprocessing. LogReg and Random Forest (RF; min_samples_leaf 2) with balanced class weights, and an MLP with early stopping [2]; seed 42. Final settings on Wed 70% / Wed 80% / Thu 70%: LogReg C = 10 / 1 / 10; RF 100 / 100 / 50 trees; MLP layers 128-64 / 64-32 / 64-32. A signed log (no fitted parameters) is followed by a StandardScaler fitted on training rows only; threshold 0.5.

Tuning protocol. The last 20% of the training rows form a time-ordered validation window. Grids: C 0.1/1/10; RF trees 50/100/200 × leaf 1/2/5; MLP 32, 64-32, 128-64. A split is tuned only if its validation window has at least 100 attacks *and* they are of a type in the fit part. Wed 70% was tuned (validation holds 10,683 FTP flows; F1 varies by at most 0.003) and Thu 70% was tuned (RF spread 0.006; all families at most 0.010). Wed 80% was not: its 69,794 validation SSH flows were never seen by the fit part (193,354 FTP flows), every setting scores validation F1 of at most 0.0009, so defaults were kept.

Metrics. Attack-class precision, recall, F1 and AP. F1 carries a 95% block-bootstrap interval (5-minute blocks, 1,000 resamples) [8]; false-positive counts carry an exact Poisson 95% interval [10]; model pairs use McNemar's test [4] (exact if b + c < 25). **False alerts per day (Task 1)** = benign flows flagged ÷ test-window hours × 24.

1.4 Task 2 configuration and evaluation

Setup. Labels were set aside first. A systematic sample of 40,000 flows per day (every 26th Wednesday, 8th Thursday flow) was signed-log transformed, scaled and reduced to 10 PCA components. **K-Means** [9]: k 2–40 × 5 seeds, silhouette [6] on 10,000 rows. **GMM**: 2–60 components (120 on Wednesday), full or diagonal covariance, BIC

[7]. **DBSCAN** [5]: 16 eps values from the k-distance curve × min_samples 5, 10, 25, 50, silhouette over settings with at most 25% noise; noise is triaged as its own cluster. A simulated analyst labels each cluster by the majority of 20 random flows; labels spent are the cost. Extra sweeps vary PCA dimensions, sample size, triage size and analyst error. Results are seed means (K-Means 5, GMM and DBSCAN 3 seeds). **Baselines:** the port 21/22 rule alerts on destination port 21 or 22; the zero-payload rule alerts when forward and backward payload bytes are both zero and duration is below 1,000 µs.

False alerts per day (Task 2) use the measured span, not 12 h. The reported value is sample false-positive rate × the day's benign flows × 24 ÷ 12; we rescale it as *reported × 12 ÷ measured span*. The flows span 9.0 h (Wednesday) and 9.5 h (Thursday), so both tasks use the traffic rate in the data. Task 1 and Task 2 false-alert numbers have different denominators and are never compared in one row.

1.5 Reproducibility

One command (*run_all.sh*) rebuilds results, figures and this PDF; *scripts/check_report_numbers.py* checks every printed number against *results/report_registry.json*. Repository: github.com/MZU-GH/SEC-635-Lab-1.

2 Results and visualizations

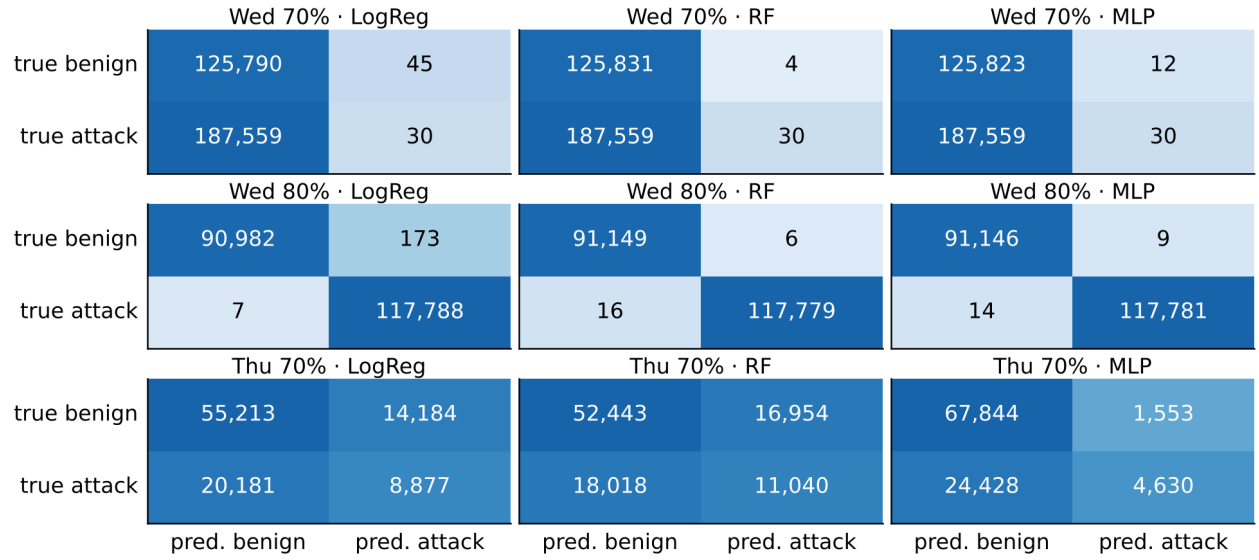


Figure 2 — RF is near-perfect on a seen campaign (6 false positives, 16 misses on Wed 80%) and unusable on Thursday. Confusion matrices on the time-ordered test windows (rows: true label; cell shading is log-scaled).

On Wed 70% every model catches the same 30 of 187,589 SSH flows. On Thu 70% RF flags 24% of benign flows (16,954) and misses 62% of attacks (18,018); the MLP trades recall (0.159) for precision.

On Wed 70% LogReg scores SSH below 0.5, yet a threshold set for 10,000 false alerts per day would reach 50.2% recall. On Thu 70% all curves approach the chance level of 0.295 (AP 0.398–0.439). On Wednesday the detectors recover both attack groups; on Thursday most infiltration flows (Figure 4, pink) sit on benign clusters, and a 2-D projection of 10 components cannot show overlap, so Table 7 measures it in full feature space.

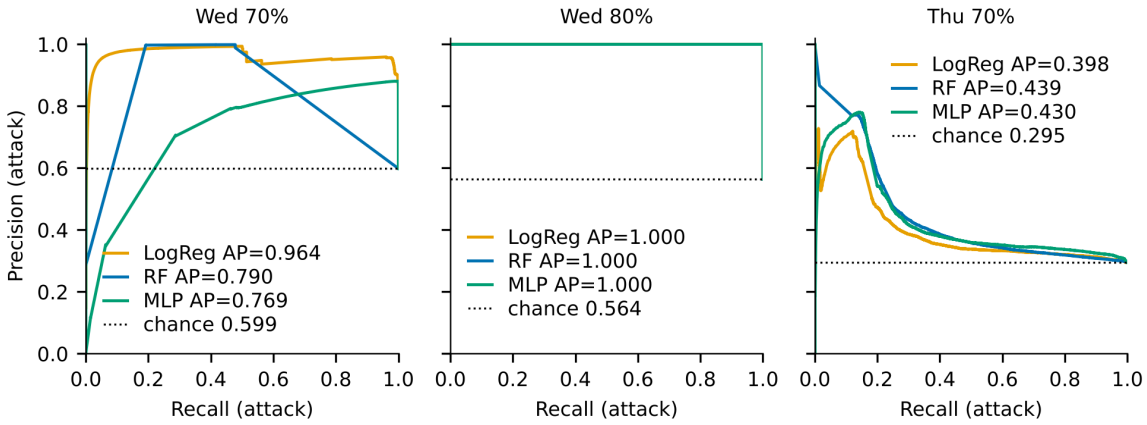


Figure 3 — LogReg ranks unseen SSH well (AP 0.964) yet its 0.5 threshold misses it: thresholds matter. Attack-class precision–recall curves; chance level (attack share) dotted.

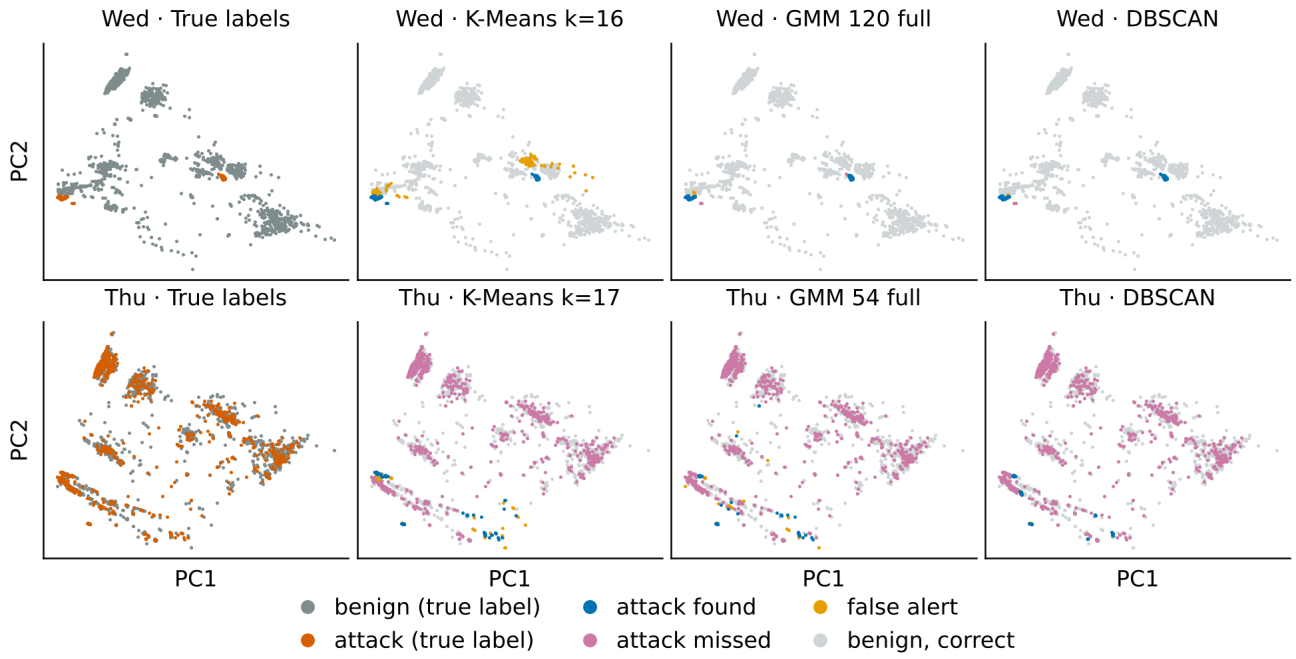


Figure 4 — Wednesday attacks form compact groups detectors isolate; Thursday attacks lie inside the benign cloud. 5,000 of the 40,000 sampled flows per day (random subsample) on PC1/PC2; left: true labels; others: detector result.

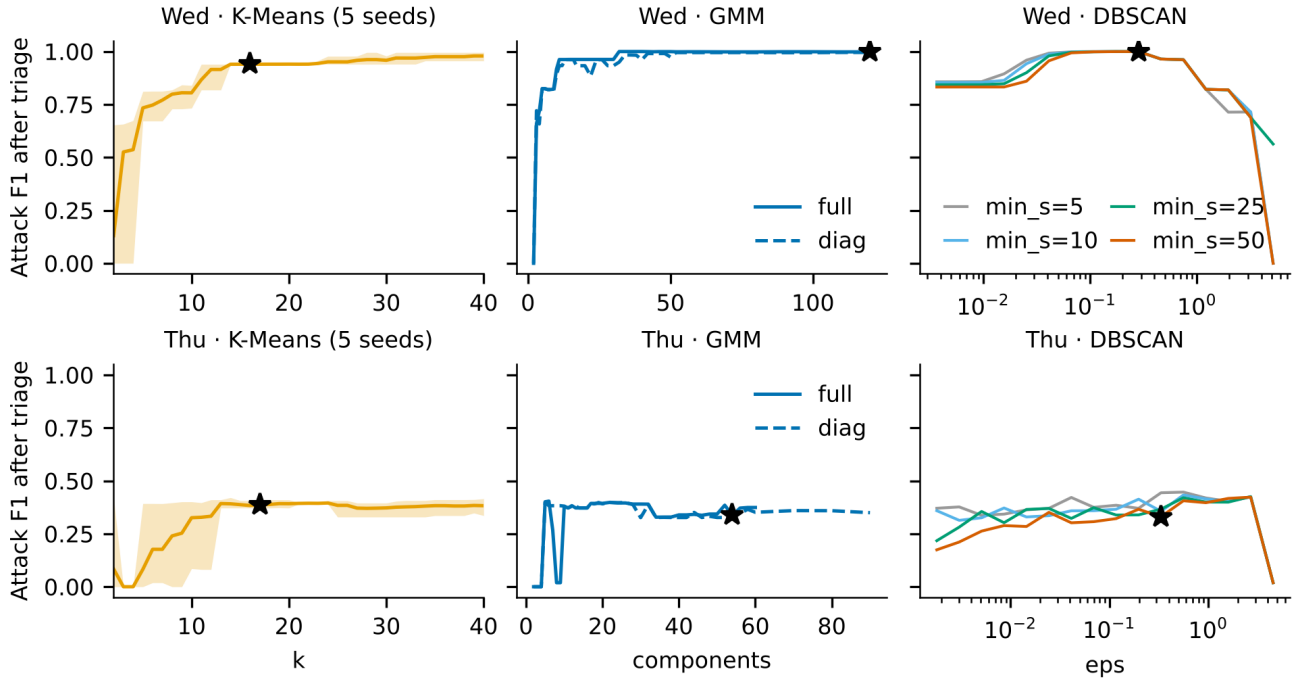


Figure 5 — Wednesday F1 sits on a wide plateau, but Thursday F1 stays low except at extreme settings, and the label-free choice (star) is rarely the best value. Attack F1 after triage per parameter (top Wednesday, bottom Thursday); K-Means band: min–max over 5 seeds; DBSCAN: one line per min_samples.

K-Means F1 flattens after the chosen k. The Wednesday GMM plateaus early, yet its BIC kept improving to the grid edge (120 components): an edge, not an optimum. DBSCAN drops at large eps on both days, where clusters merge, and on Thursday no setting exceeds F1 0.447.

3 Metric comparisons

3.1 Task 1: supervised classifiers

Table 3 — Task 1, attack class, time-ordered test windows. FP = benign flows flagged. False alerts/day = FP ÷ test hours × 24, with exact Poisson 95% CI. Bold: best F1 and fewest false alerts per split. Rules have no score, so no AP.

Split / model	Precision	Recall	F1	F1 95% CI	AP	FP	False alerts/day [95% CI]
Wed 70%							
LogReg	0.4000	0.0002	0.0003	[0.000, 0.0012]	0.964	45	306 [223–410]
RF	0.8824	0.0002	0.0003	[0.000, 0.0012]	0.790	4	**27 [7–70]**
MLP	0.7143	0.0002	0.0003	[0.000, 0.0012]	0.769	12	82 [42–143]
Port 21/22 rule	0.9997	1.000	**0.9998**	–	–	63	429 [330–549]
Wed 80%							
LogReg	0.99853	0.99994	0.99924	[0.99907, 0.99935]	0.99998	173	1,421 [1,217–1,650]
RF	0.99995	0.99986	**0.99991**	[0.99979, 0.99996]	>0.99999	6	**49 [18–107]**
MLP	0.99992	0.99988	**0.99990**	[0.99980, 0.99996]	>0.99999	9	74 [34–140]
Port 21/22 rule	0.99963	1.000	0.99981	–	–	44	361 [263–485]
Thu 70%							
LogReg	0.385	0.305	0.341	[0.174, 0.462]	0.398	14,184	110,177 [108,371–112,005]
RF	0.394	0.380	**0.387**	[0.231, 0.504]	0.439	16,954	131,693 [129,718–133,691]
MLP	0.749	0.159	0.263	[0.040, 0.443]	0.430	1,553	12,063 [11,471–12,678]
Port 21/22 rule	0.286	0.008	0.015	–	–	565	**4,389 [4,034–4,766]**
Zero-payload rule	0.379	0.229	0.286	–	–	10,943	85,002 [83,417–86,610]

Wednesday. On Wed 70% the models are equally blind (recall 0.0002) while the port rule scores F1 0.9998. On Wed 80% RF and MLP are indistinguishable (F1 0.99991, 0.99990), the port rule matches them on detection (F1 0.99981), and RF's measured advantage is false alerts: 49 per day against 361, with non-overlapping intervals. **Thursday.** No model works: RF F1 0.387, LogReg 0.341, MLP 0.263, at 12,063–131,693 false alerts per day; the port rule catches 0.8% of infiltration and the zero-payload rule (F1 0.286) is no better than the best classifier.

Table 4 — Does the split matter? F1 under a random split, the honest time split and (Thursday) the unrepaired clock, run for RF only. Thursday time-split cells carry the 95% block-bootstrap CI.

Evaluation	LogReg	RF	MLP
Wed: random split, 70% train	0.99987	0.99997	0.99997
Wed: honest time split (Wed 80%)	0.99924	0.99991	0.99990
Thu: random split, 70% train	0.438	0.489	0.389
Thu: honest time split (Thu 70%)	0.341 [0.174, 0.462]	0.387 [0.231, 0.504]	0.263 [0.040, 0.443]
Thu: unrepaired clock, 70% train	not run	0.592	not run

The split matters little on Wednesday and moderately on Thursday; the clock matters most. On Wednesday random-split inflation is negligible (F1 differs by at most 0.0006) because brute force is trivially separable. On Thursday all three models score higher on a random split (LogReg +0.097, RF +0.102, MLP +0.127), yet each random value lies inside its time-split CI, so the direction is consistent but the size is uncertain. The unrepaired clock is the largest distortion: Thursday RF scores 0.592, outside the time-split CI [0.231, 0.504]. The earlier comparison against Wed 70% mixed leakage with an unseen attack type (RF F1 0.0003) and is withdrawn.

Significance (McNemar; b = flows only the first model gets right, c = only the second). Wed 80%: RF vs MLP is a tie (b = 5, c = 4, p = 1.00); both beat LogReg ($p < 0.001$); RF beats the port rule ($p = 0.01$), with 22 errors against 44. On Wed 70% pairs differ only through a few false positives ($p < 0.001$). Thu 70%: every pair differs ($p < 0.001$, $p < 0.001$, $p < 0.001$ for RF vs the port rule), but McNemar treats flows as independent and the block-bootstrap CIs overlap widely (RF [0.231, 0.504], MLP [0.040, 0.443]), so the ranking is weak evidence.

3.2 Task 2: unsupervised detectors

Table 5 — Task 2 on the 40,000-flow samples, detectors chosen without labels. F1 is the seed mean, with its min–max over seeds. False alerts/day use the measured-span convention (Section 1.4). Bold: best F1 and fewest false alerts per day.

Day / method	Chosen setting	Precision	Recall	F1	F1 seed range	Analyst labels	False alerts/day	At edge?
Wednesday								
K-Means	k 16	0.8862	1.000	0.9397	0.9392–0.9401	320	129,774	interior
GMM	120, full	0.9988	0.9972	0.9980	0.9980–0.9981	2,218	1,178	upper edge
DBSCAN	eps 0.284, ms 50, 34 cl.	0.9997	0.9992	0.9994	0.9994–0.9994	700	**347**	ms at edge
Port rule		0.9990	1.000	**0.9995**	–	0	970	–
Thursday								
K-Means	k 17	0.5936	0.2934	0.3875	0.3718–0.4063	340	52,642	interior
GMM	54, full	0.8163	0.2311	0.3546	0.3357–0.3906	1,080	15,902	interior
DBSCAN	eps 0.333, ms 50, 52 cl.	0.8310	0.2243	0.3470	0.3239–0.3864	1,060	14,749	ms at edge
Port rule		0.3574	0.0094	0.0184	–	0	**4,112**	–
Zero-payload rule		0.4668	0.3360	**0.3907**	–	0	93,065	–

Wednesday. All three detectors separate brute force; DBSCAN (F1 0.9994) matches the port rule (F1 0.9995) at fewer false alerts per day (347 against 970) for 700 analyst labels, while K-Means raises 129,774 per day (F1 0.940). **Thursday.** F1 is 0.388 (K-Means), 0.355 (GMM) and 0.347 (DBSCAN); the zero-payload rule scores 0.3907 on the same sample with no labels.

Table 6 — Stability: seed-mean attack F1 range (min–max). Local = within one grid step of the choice; PCA 5–20 components (and 11/13 for 95% variance); sample 10,000–80,000 flows; triage 5–50 flows per cluster; analyst error 0%–10%.

Day	Detector	Local (±1 step)	PCA dims	Sample size	Triage n	Analyst error
Wed	K-Means	0.9396–0.9397	0.9331–0.9490	0.9398–0.9424	0.9398	0.9398
Wed	GMM	0.9980	0.9980–0.9987	0.9976–0.9986	0.9980	0.9980
Wed	DBSCAN	0.9640–0.9994	0.9638–0.9996	0.9620–0.9994	0.9994	0.9994
Thu	K-Means	0.384–0.388	0.318–0.398	0.384–0.398	0.294–0.398	0.398–0.408
Thu	GMM	0.337–0.385	0.355–0.369	0.355–0.393	0.355–0.380	0.345–0.355
Thu	DBSCAN	0.331–0.406	0.323–0.380	0.333–0.366	0.328–0.366	0.347–0.354

Stability. On Thursday the detectors lie within 0.041 F1 and their seed ranges overlap (0.372–0.386), so they cannot be ranked. F1 moves most with triage size for K-Means (0.104), PCA dimension for DBSCAN (0.057) and sample size for GMM (0.038); analyst error up to 10% moves it by at most 0.010. Only 29 of 64 Thursday DBSCAN settings meet the noise cap, and F1 spans 0.019–0.447.

4 Critical reflection

4.1 Where each technique added value and fell short

Supervised, value. RF is near-perfect on a campaign it has seen: on Wed 80% F1 0.99991 at 49 false alerts per day, the lowest point estimate there (LogReg 1,421, port rule 361; the MLP's 74 [34–140] overlaps). Brute-force flows are easy to separate: the median found brute-force flow has 333,333 packets/s and a 26,883-byte initial window, against 111 and 278 for benign. **Shortfall.** RF is blind to an unseen attack type (recall 0.0002 on Wed 70%; the port rule 1.000). On infiltration it flags 24% of benign traffic at threshold 0.5, and a threshold chosen on validation for at most 1,000 false alerts per day gives 12,009 on test (12× the budget, recall 15%) against 981 on validation: the budget does not transfer between windows (distribution shift).

Unsupervised, value. Wednesday brute force is separated without labels: DBSCAN reaches F1 0.9994 at 347 false alerts per day for 700 analyst labels, GMM F1 0.9980. **Shortfall.** The selection criteria are unstable at the grid edges. GMM BIC never turns by 120 components on Wednesday. DBSCAN silhouette keeps rising past min_samples 50: it goes from 0.817 to 0.876 (min_samples 200) on Wednesday and from 0.751 to 0.758 (100) on Thursday, while F1 falls from 0.999 to 0.964 and from 0.331 to 0.299. The label-free choice is not settled.

4.2 What was found and what was missed (infiltration)

Table 7 compares the groups. **(a) Most clustering finds are reproducible by one line.** The zero-payload rule catches 2,923 of the 3,482 flows any detector found (84%); the port rule catches 54. **(b) The interesting finds are the single-algorithm ones no rule reproduces.** The GMM-only (85) and DBSCAN-only (97) groups are long, near-silent connections (median 687,489 and 9,014,173 µs); the payload rule catches 1 and 0 of them, the port rule 8 and 0. **(c) Missed flows look like benign traffic.** 86% of the 8,076 flows missed by all have mostly benign neighbours, against 7% of found flows; their top ports are 53, 443 and 3389 (39%, 18%, 14%). **(d) Label noise caps any flow-level detector.** 40% of all sampled infiltration flows (80% of the found ones, 23% of the missed) are feature-identical to a

benign flow, because infiltration labels are assigned by host and time window [3]. It does not separate found from missed but explains why found clusters carry false alerts.

Table 7 — Thursday infiltration in the 40,000-flow sample: found versus missed. Medians per group (B = total payload bytes). Payload and port rule: flows caught (share of group). Identical to benign: all model features equal to a benign flow anywhere in the day. Benign neighbours: more than 5 of 10 nearest PCA-space neighbours in the sample are benign.

Group	Flows	Dur. (µs)	Fwd / bwd B	Pkts/s	Init. win.	Payload rule	Port rule	Identical to benign	Benign nbrs
Benign (reference)	28,442	18,684	43 / 89	157	256	4,436 (16%)	196 (1%)	–	89%
Found by any detector	3,482	250	0 / 0	8,000	1,024	2,923 (84%)	54 (2%)	80%	7%
K-Means only	984	2	0 / 0	1,000,000	1,024	931 (95%)	2 (0%)	–	–
GMM only	85	687,489	0 / 0	4.9	128	1 (1%)	8 (9%)	–	–
DBSCAN only	97	9,014,173	0 / 0	0.3	8,192	0 (0%)	0 (0%)	–	–
Missed by all	8,076	17,882	46 / 105	158	237	960 (12%)	55 (1%)	23%	86%
All infiltration (sampled)	11,558	950	35 / 48	2,161	1,024	3,883 (34%)	109 (1%)	40%	62%

Not computed per single-detector group (–); for benign flows the first test is trivially true.

4.3 Limits

The analyst is assumed error-free in the headline results (error of 0%–10% moves F1 by at most 0.010). Daily counts assume a constant rate over 9.0 h and 9.5 h spans. The port rule was written with hindsight. The GMM grid uses a single seed; each sample is 40,000 flows. Wed 80% is an optimistic bound. The attack schedule was not verified externally.

5 Production recommendation

Random Forest on Wednesday-like brute force: this model raises 49 false alerts per day (95% CI 18–107), and here is why that is acceptable: at 1 minute per alert it costs 0.8 analyst-hours per day (10% of one 8-hour shift; 1.8 hours at the upper CI bound) while it catches 99.99% of a campaign it has seen.

Random Forest on Thursday infiltration: this model raises 131,693 false alerts per day, and here is why that is not acceptable: that is 2,195 analyst-hours per day, or 274 analysts on 8-hour shifts, and it still catches only 38.0% of infiltration flows. A validation-chosen threshold still gives 12,009 per day at 15% recall.

Table 8 — What to deploy and what it costs. False alerts/day: Task 1 convention (supervised, port rule) or Task 2 measured-span convention (clustering, payload rule); not comparable across rows.

Technique	Deploy as	Measured cost
Port 21/22 rule	FTP/SSH brute-force alerting.	361–429 false alerts/day (Wednesday windows); recall 0.8% on infiltration; no labels.
Random Forest	Second-line scoring, labelled families.	49 false alerts/day on Wed 80% (312 fewer than the rule); recall 0.0002 on unseen attacks; 92 s training.
GMM / DBSCAN	Daily hunting review, not alerting.	700–2,218 labels per 40,000 flows; 1,178–15,902 (GMM), 347–14,749 (DBSCAN) false alerts/day if alerting.
Zero-payload rule	Hunting query, not an alert.	91,446 false alerts/day at recall 33%, precision 0.461.
Not deployed	LogReg, MLP, K-Means.	LogReg (1,421 false alerts/day, 29× RF); MLP (ties RF, p = 1.00; recall 0.159 on Thursday); K-Means (129,774 false alerts/day on Wednesday).

Do this. Fix the clock at ingestion. Retrain on a rolling, time-ordered schedule and keep a rule for unlabelled attack types. Move infiltration detection to host and session level: flow features carry no signal for 40% of it.

Repository: github.com/MZU-GH/SEC-635-Lab-1

References

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